

ECEN 607: Advanced Analog Circuit Tech

Lab 2: MOS Two stage Amplifier

Name: Sheena Selvakumar

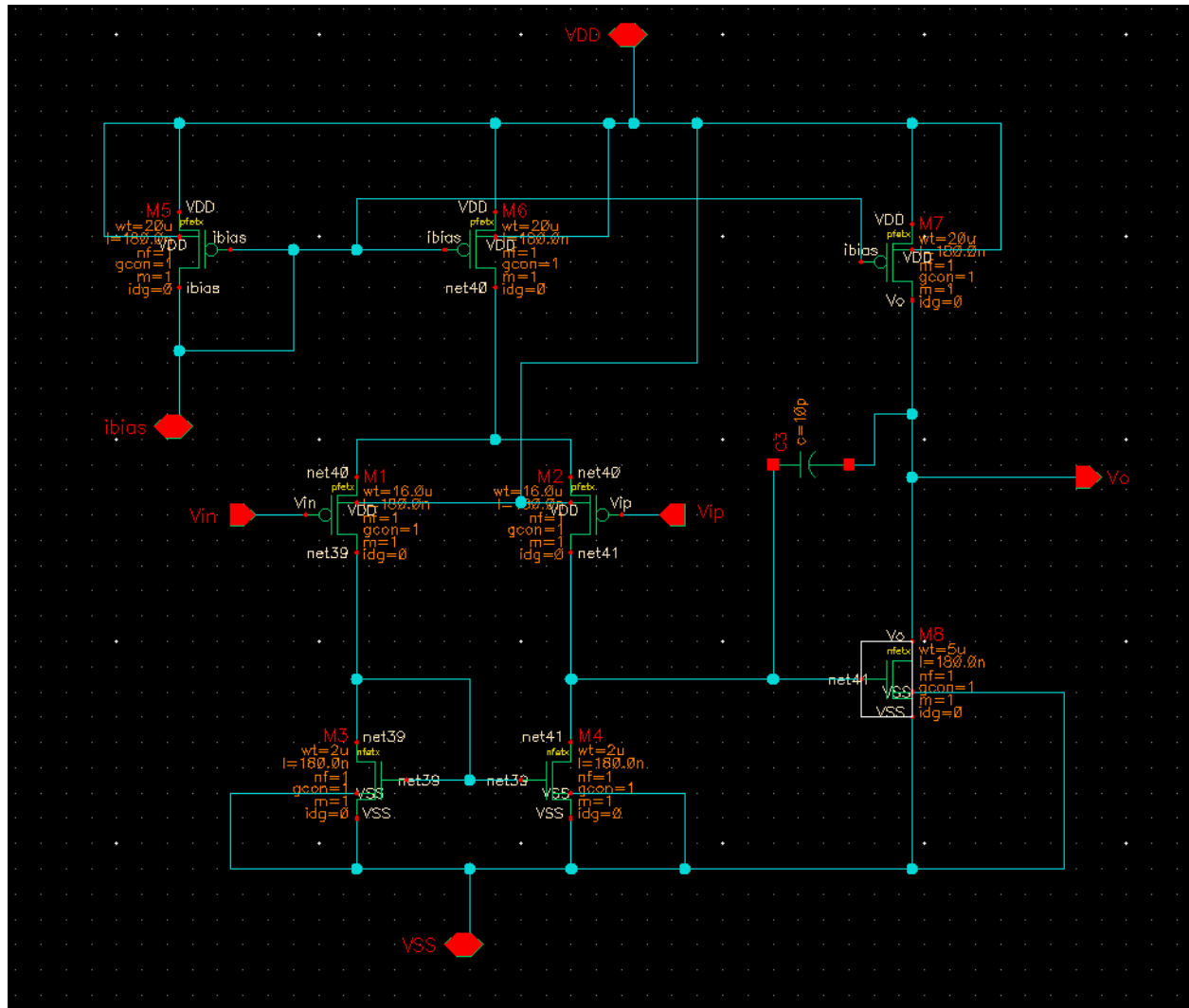
UIN: 537000654

Pre-lab or Lab Report

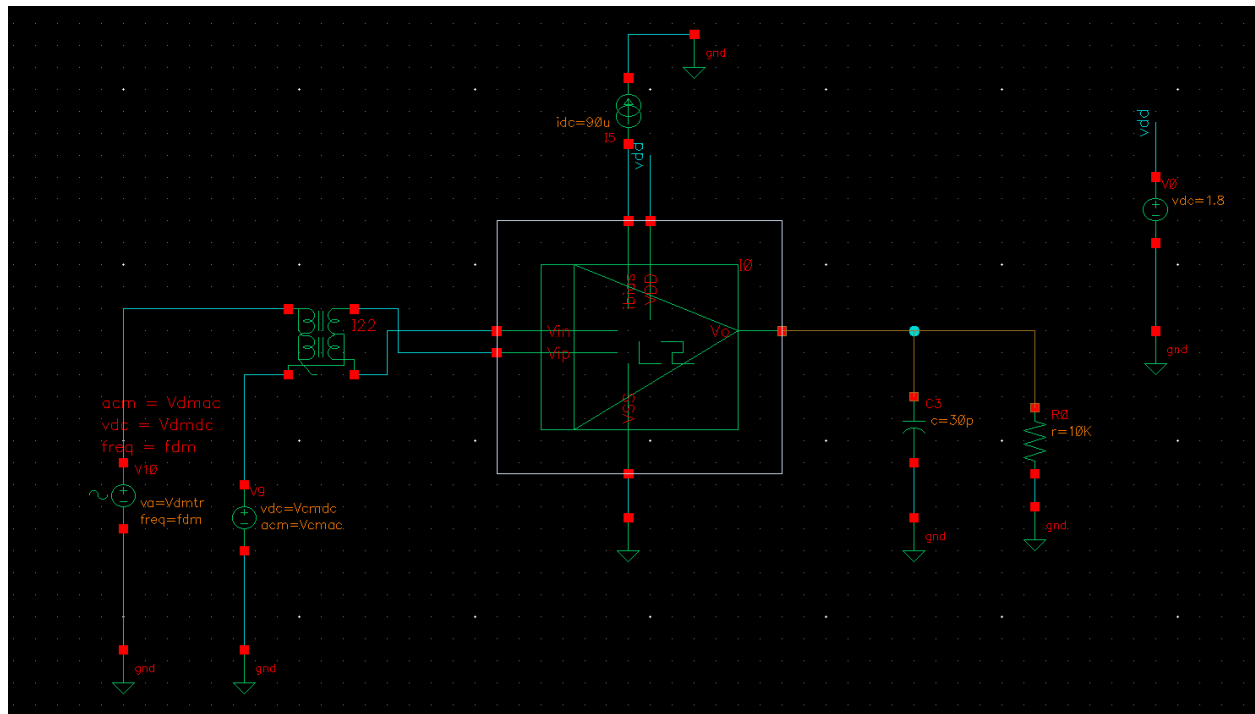
Lab Report: Provide necessary plots, explanations etc. in the report for the following questions.

1. Simulate the design from the prelab and adjust the transistor sizes accordingly until all specifications are met.

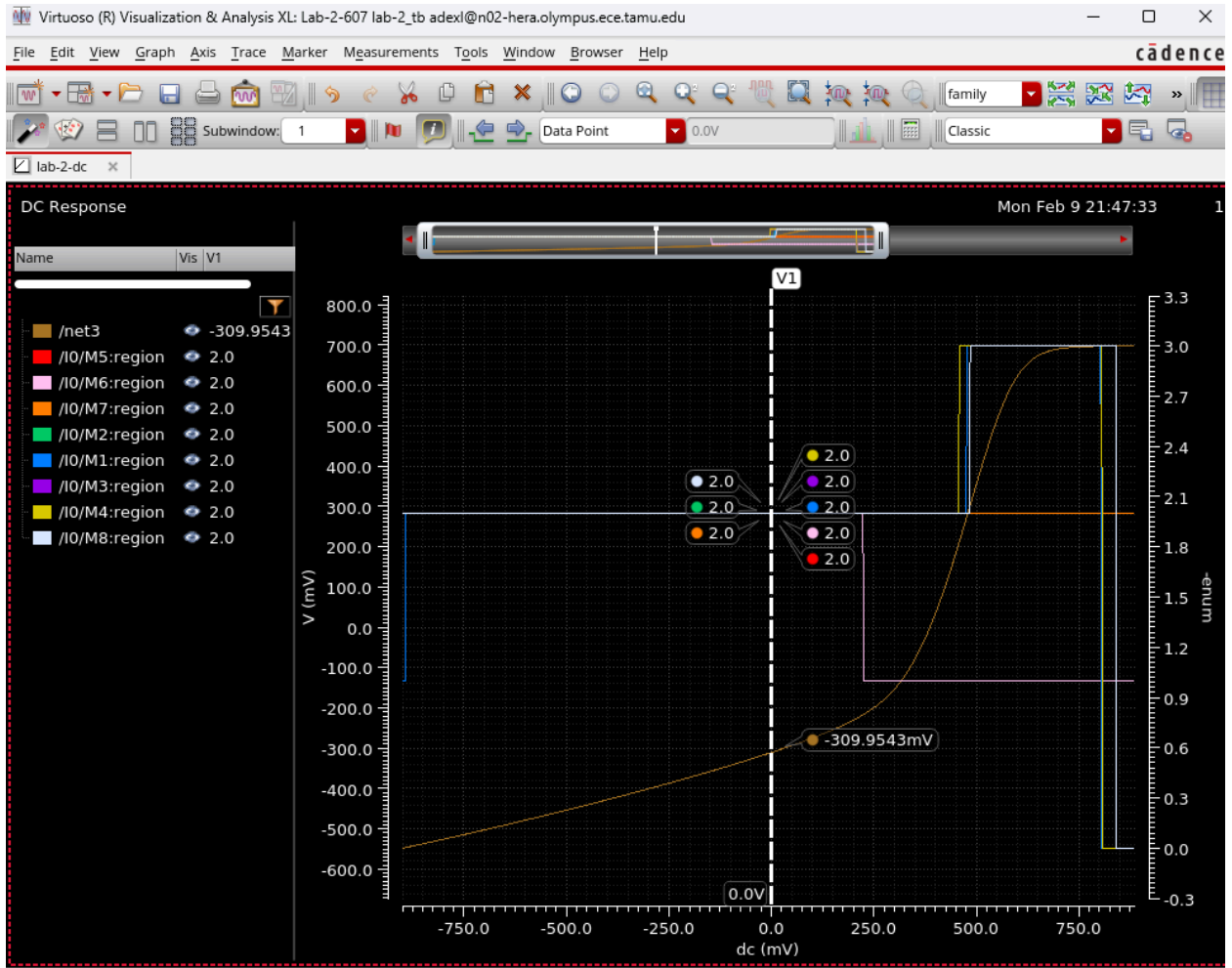
Schematic:

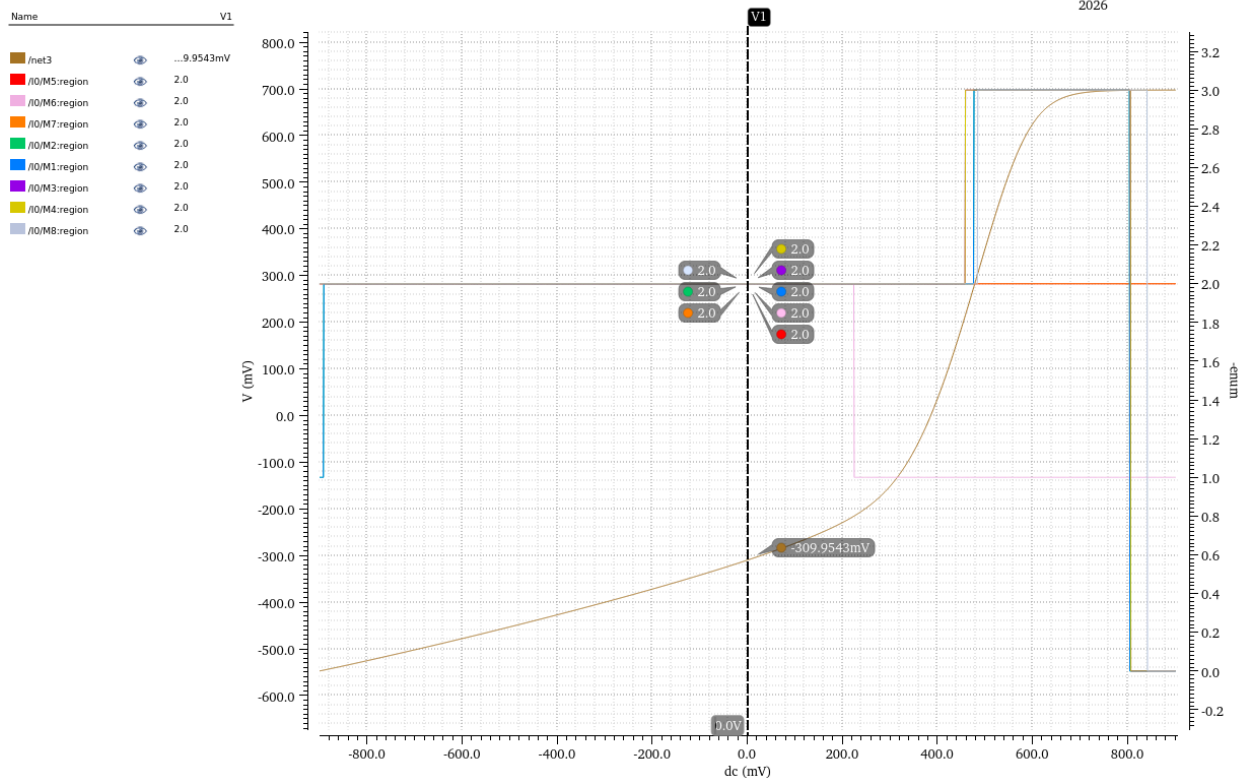


Test bench setup:

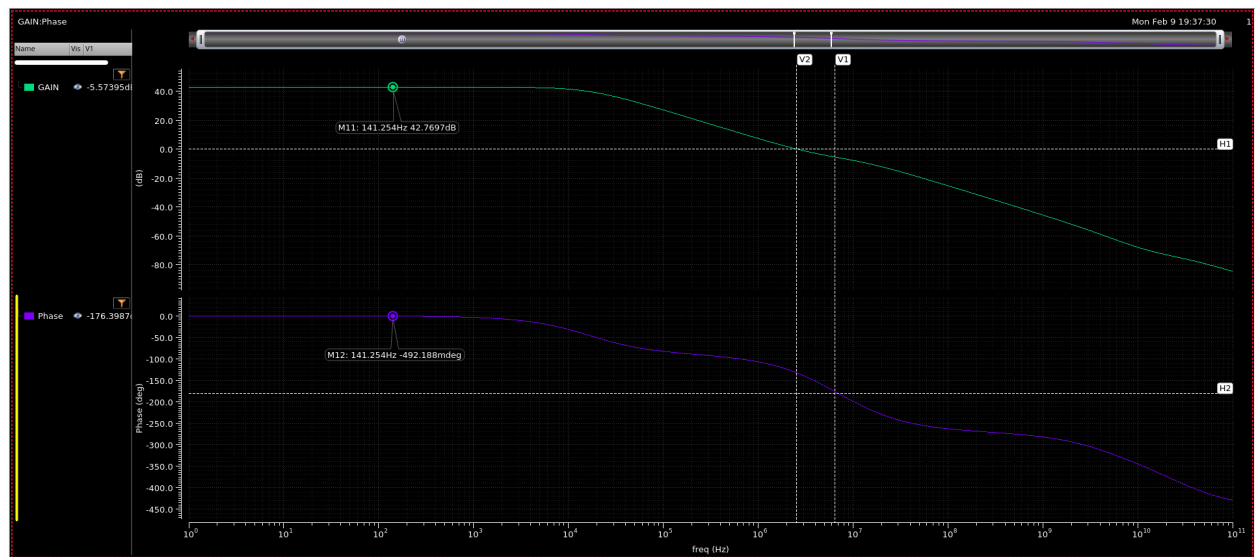


Output dc analysis:

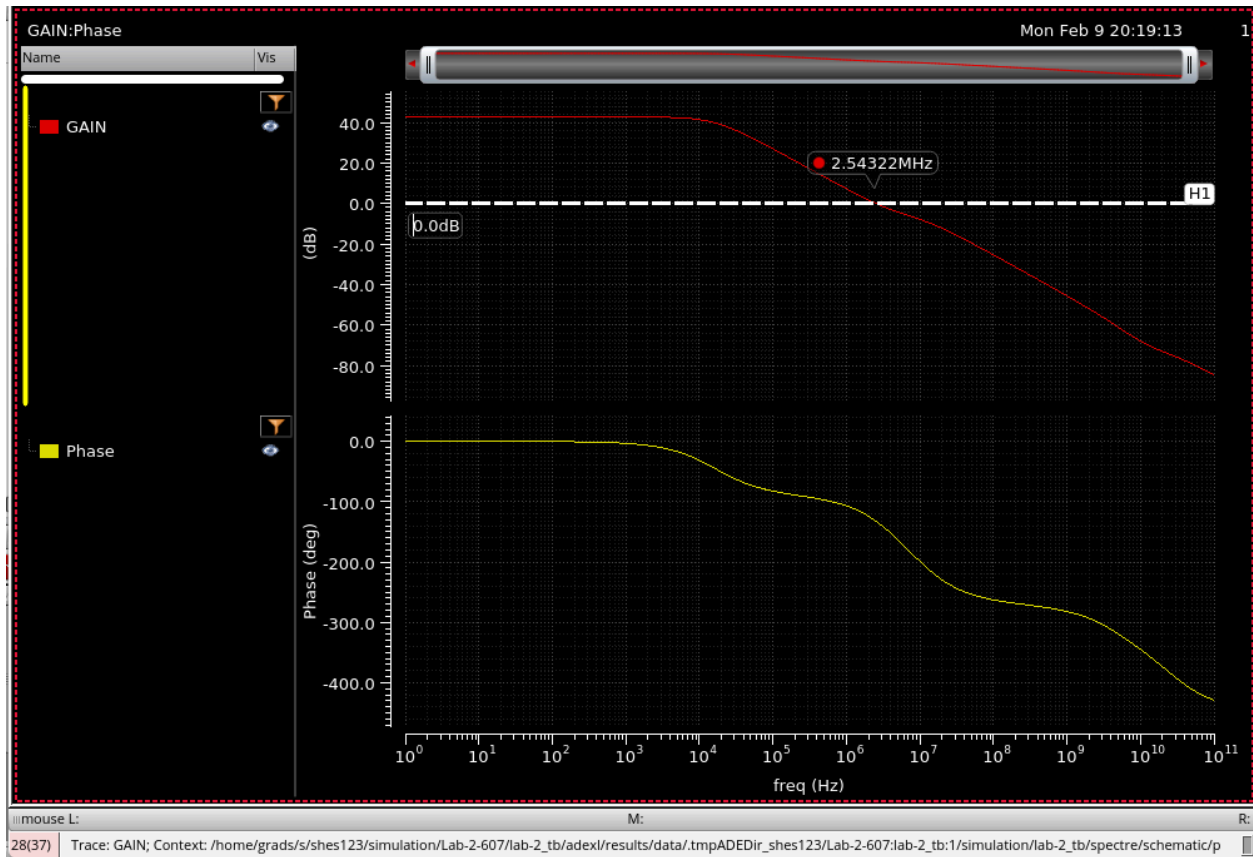




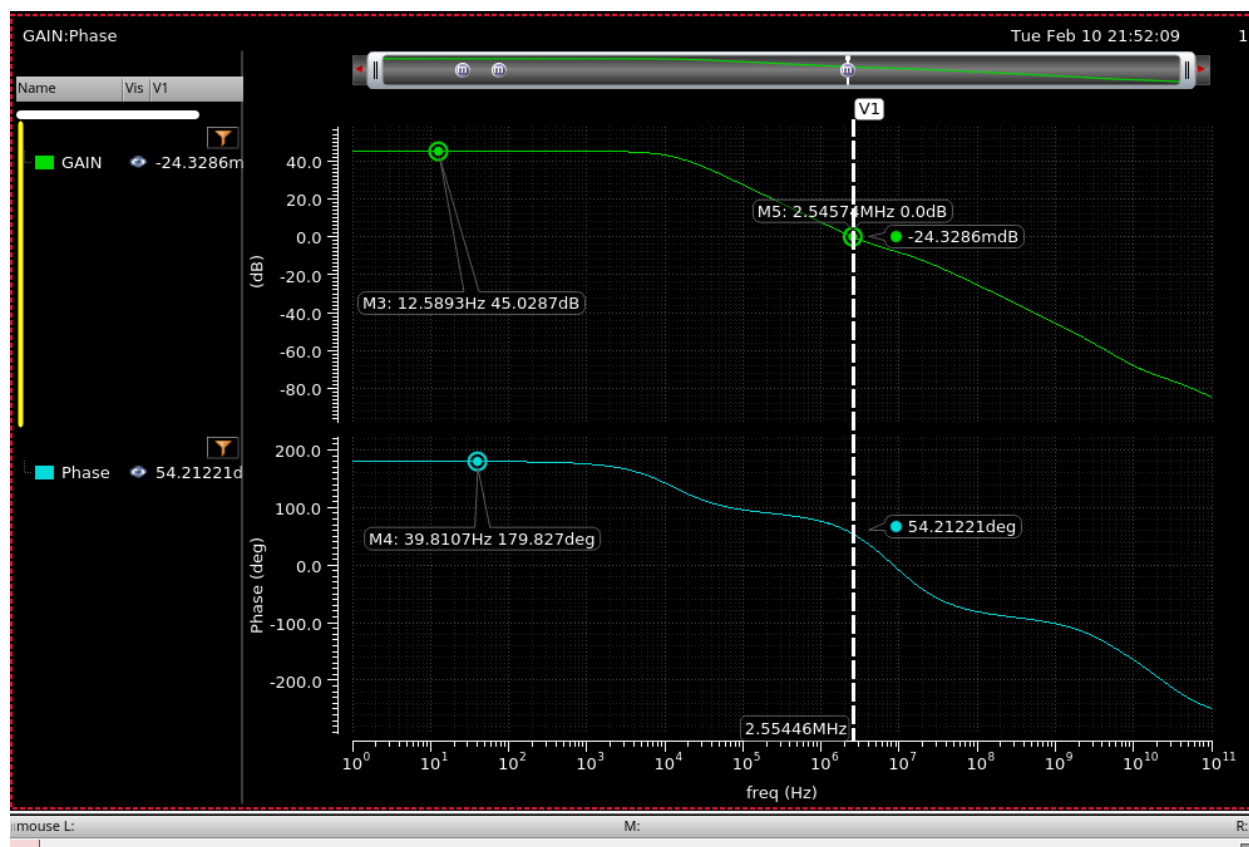
Output ac analysis(gain,phase margin and gain margin)

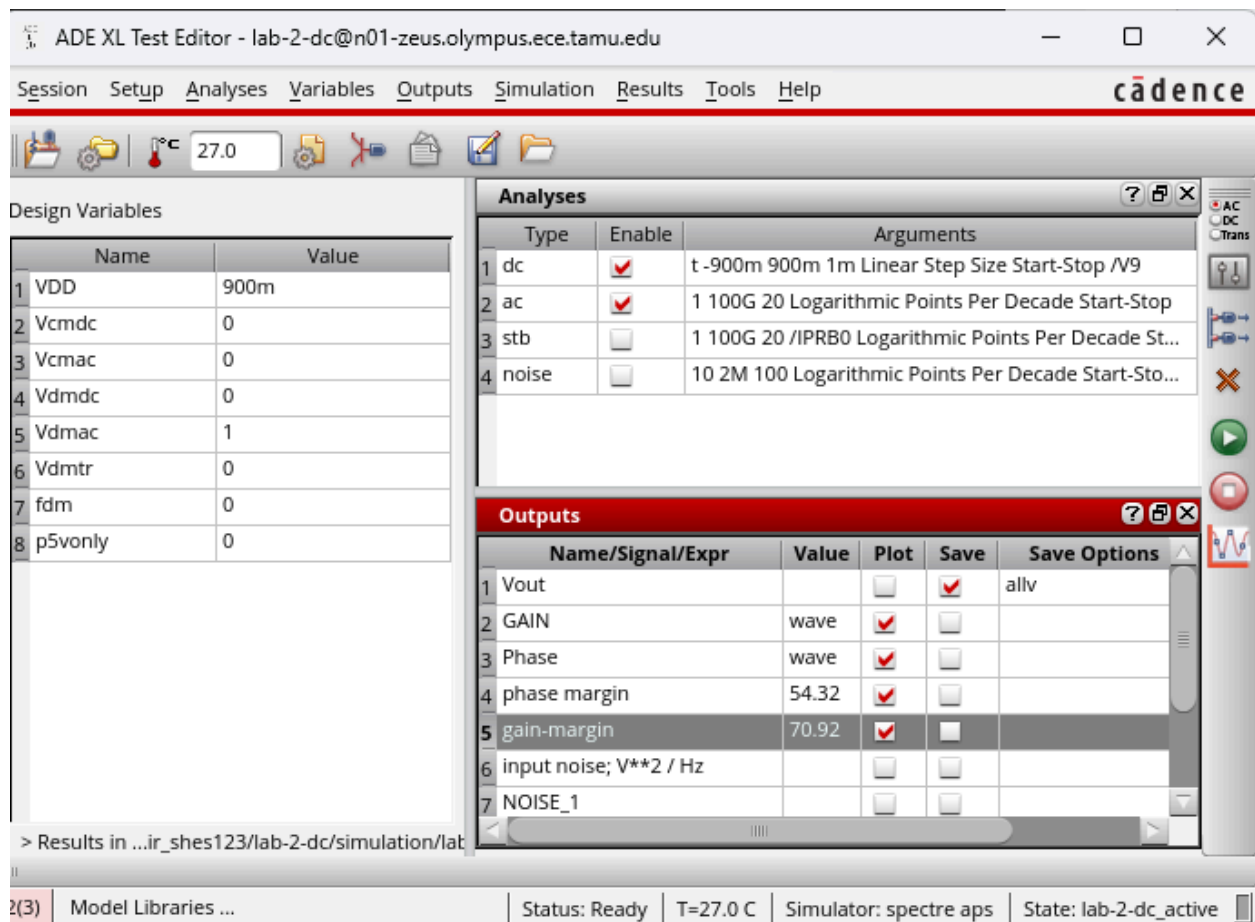


GBW:



Phase_margin satisfied_54.32 degree:



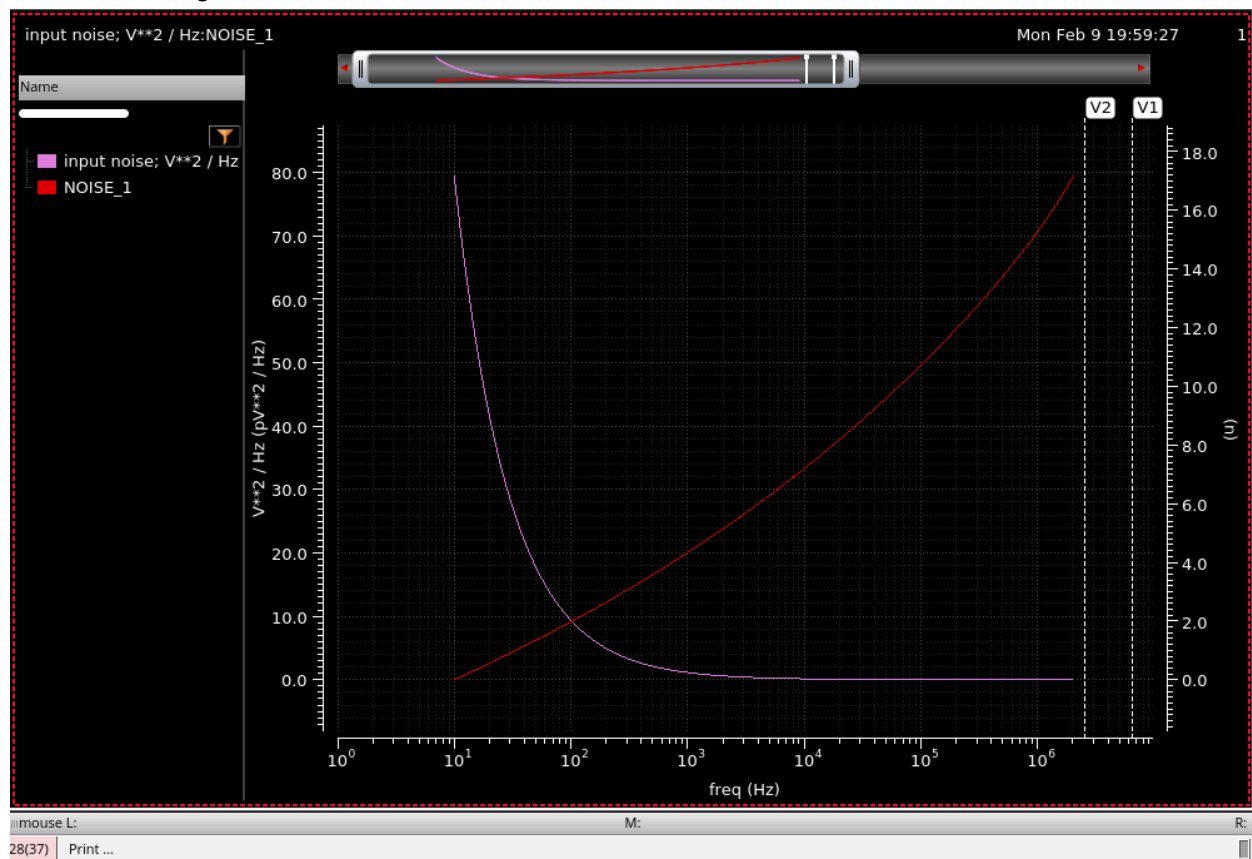


The design is simulated using a schematic and test bench setup. Transistor sizes are adjusted through simulation to meet all specifications. Initially, hand-calculated values for transistor width (W) and length (L) are provided, but after running simulations, these values are modified to ensure the circuit meets required parameters like gain, phase margin, and bandwidth. For example, M1 and M2 were originally sized at 1u/180nm but increased to 16u/180nm after simulation to meet specifications

2. Report the input referred RMS voltage noise integrated from (10 Hz – 2 MHz)

According to the lab manual, noise analysis and integrated noise are performed to obtain input referred RMS voltage noise. The details and exact value should be reported based on your simulation results as shown in the "Noise and Integrated noise" section

Noise and Integrated noise:



3. In addition, provide comparison tables of hand-calculated vs. final transistor sizes, and required

A comparison table is included showing hand-calculated vs. simulation-based transistor sizes. The simulation results show substantial size adjustments to meet specs. For example, M8 changed from 1u to 15u. Simulated specs vs. required specs are summarized in another table, showing values like $AV_0 = 45.03$ dB, $GBW = 2.54$ MHz, $PM = 54.32$ degrees, Output swing = 0.601 V, $CL = 30$ pF, $RL = 10$ k Ω , and $I_{bias} = 90$ uA. The manual suggests commenting on any deviations and improvements brought by simulation compared to theoretical calculations

Table containing hand calculated values of W and L values of transistors

Transistor Number	Width	Length(nm)
M1	1u	180
M2	1u	180
M3	2u	180

Transistor Number	Width	Length(nm)
M1	1u	180
M2	1u	180
M4	2u	180
M5	19u	180
M6	19u	180
M7	19u	180
M8	1u	180

Transistor sized on doing the simulation:

Transistor Number	Width	Length(nm)
M1	16u	180
M2	16u	180
M3	4u	180
M4	4u	180
M5	20u	180
M6	20u	180
M7	20u	180
M8	15u	180

Comments:

specs (Table 1-1) vs. simulated specs. Comment on your results.

Table for spec obtained from simulating the circuit:

VDD	0.9
VSS	-0.9

AV0	45.03.db
GBW	2.54MHz
PM	54.32 degrees
Output swing	0.601 V
CL	30pF
RL	10kohm
cc	60pF
ibias	90u

4. Run a large Monte Carlo (MC) simulation on the op-amp design. Be sure to run this simulation

with process variation and mismatch. Generate histograms of the following parameters:

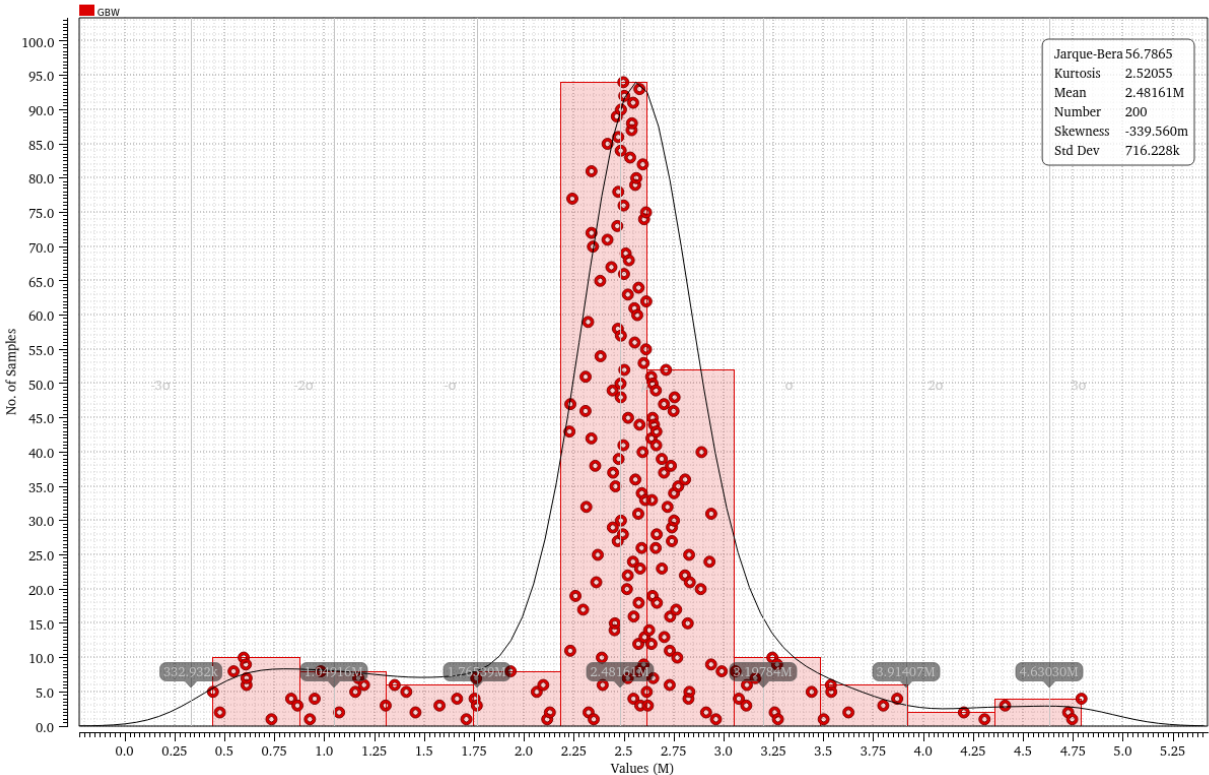
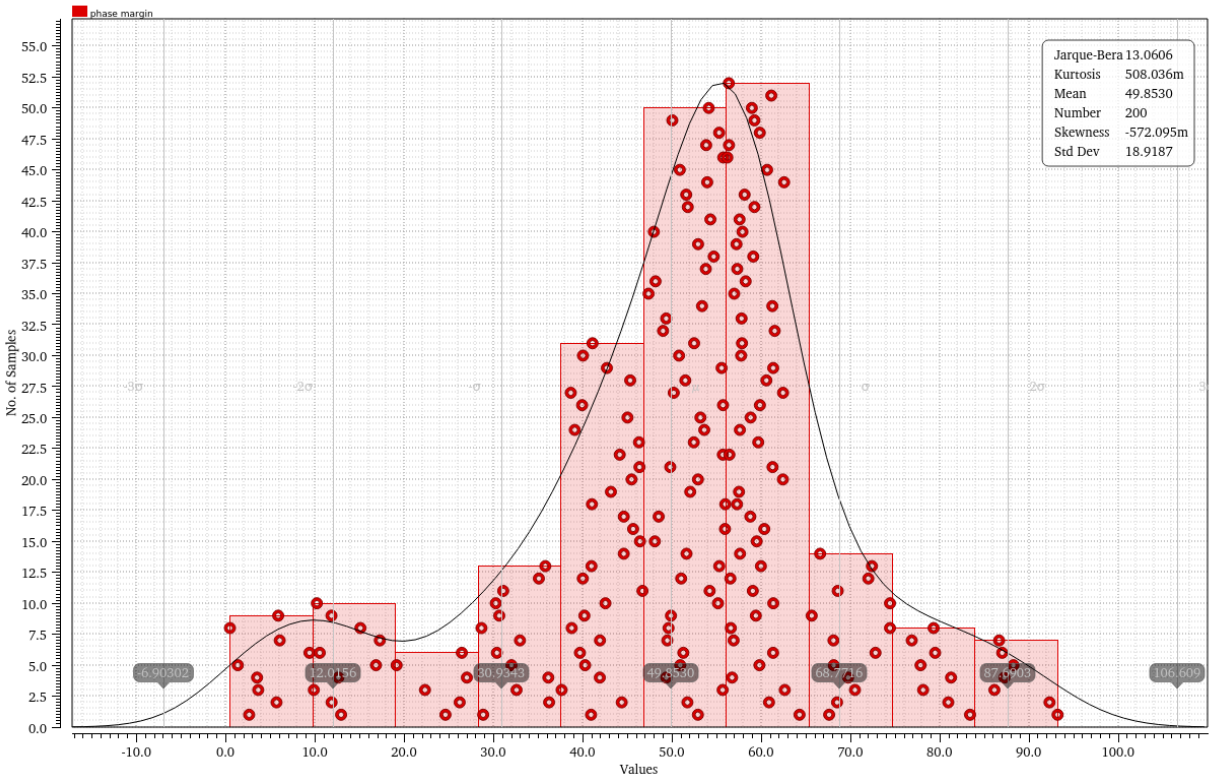
- a) DC Gain: Av0
- b) Gain-bandwidth product: GBW
- c) Phase margin : PM

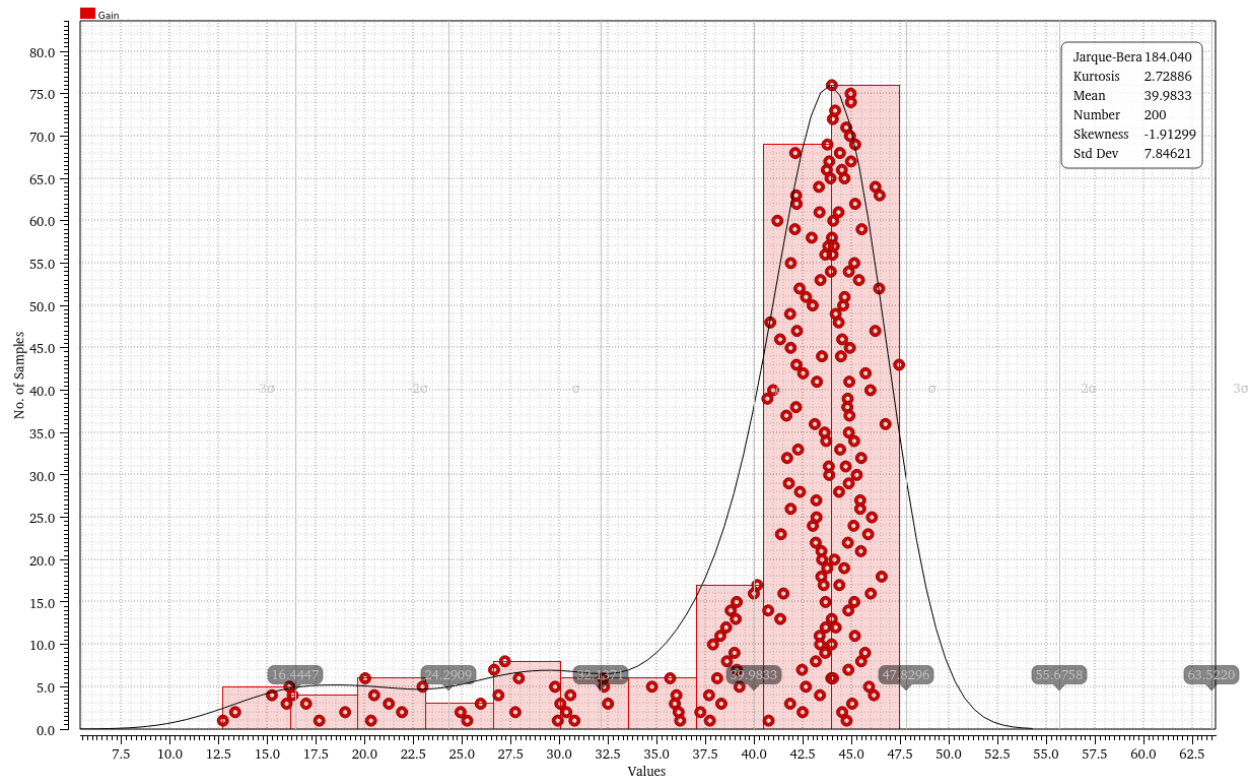
MC simulation is conducted on the op-amp, including process variations and mismatch.

Histograms are generated for:

- a) DC Gain (Av0)
- b) Gain-bandwidth product (GBW)
- c) Phase margin (PM)

This helps to statistically analyze the parameter spread and identify if the design meets specs in all possible manufacturing cases

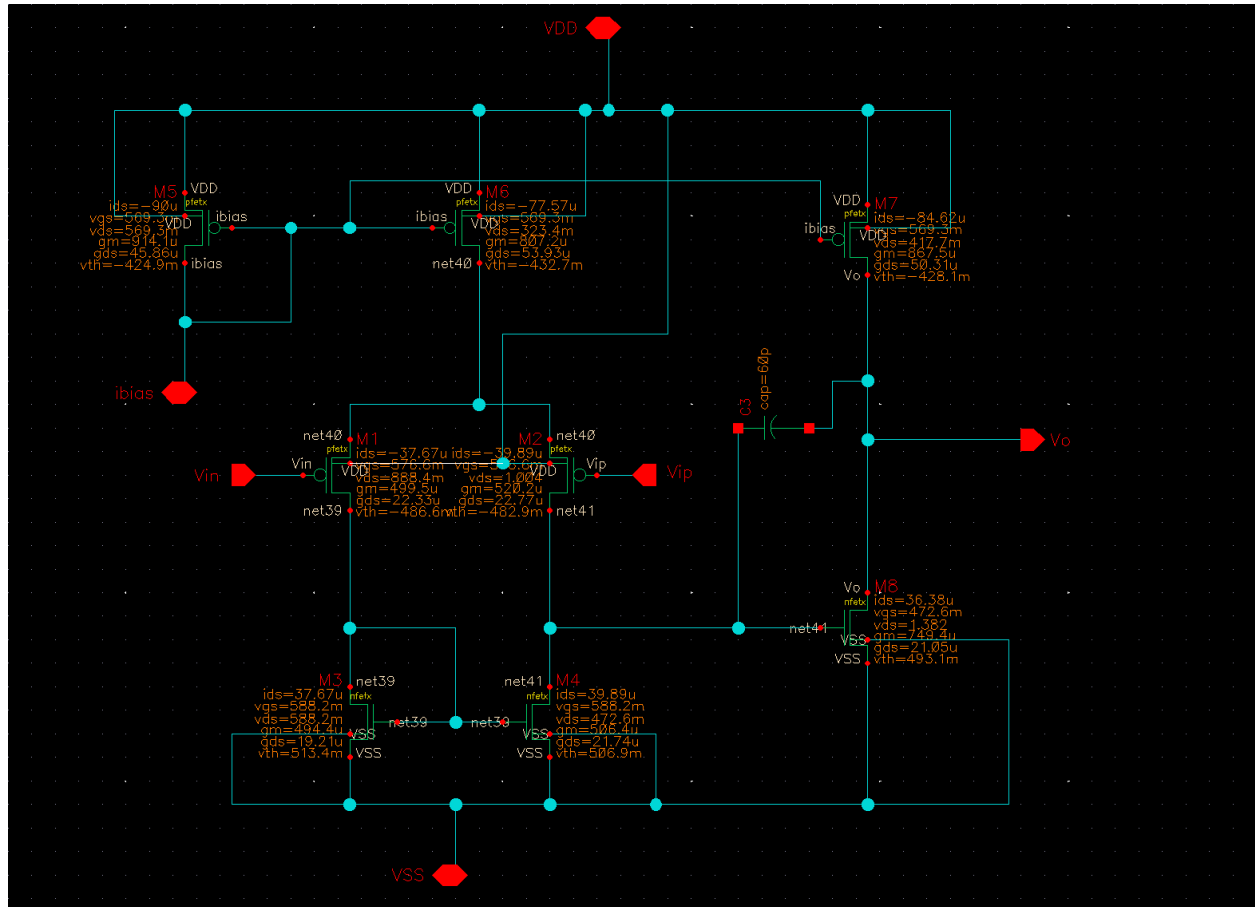




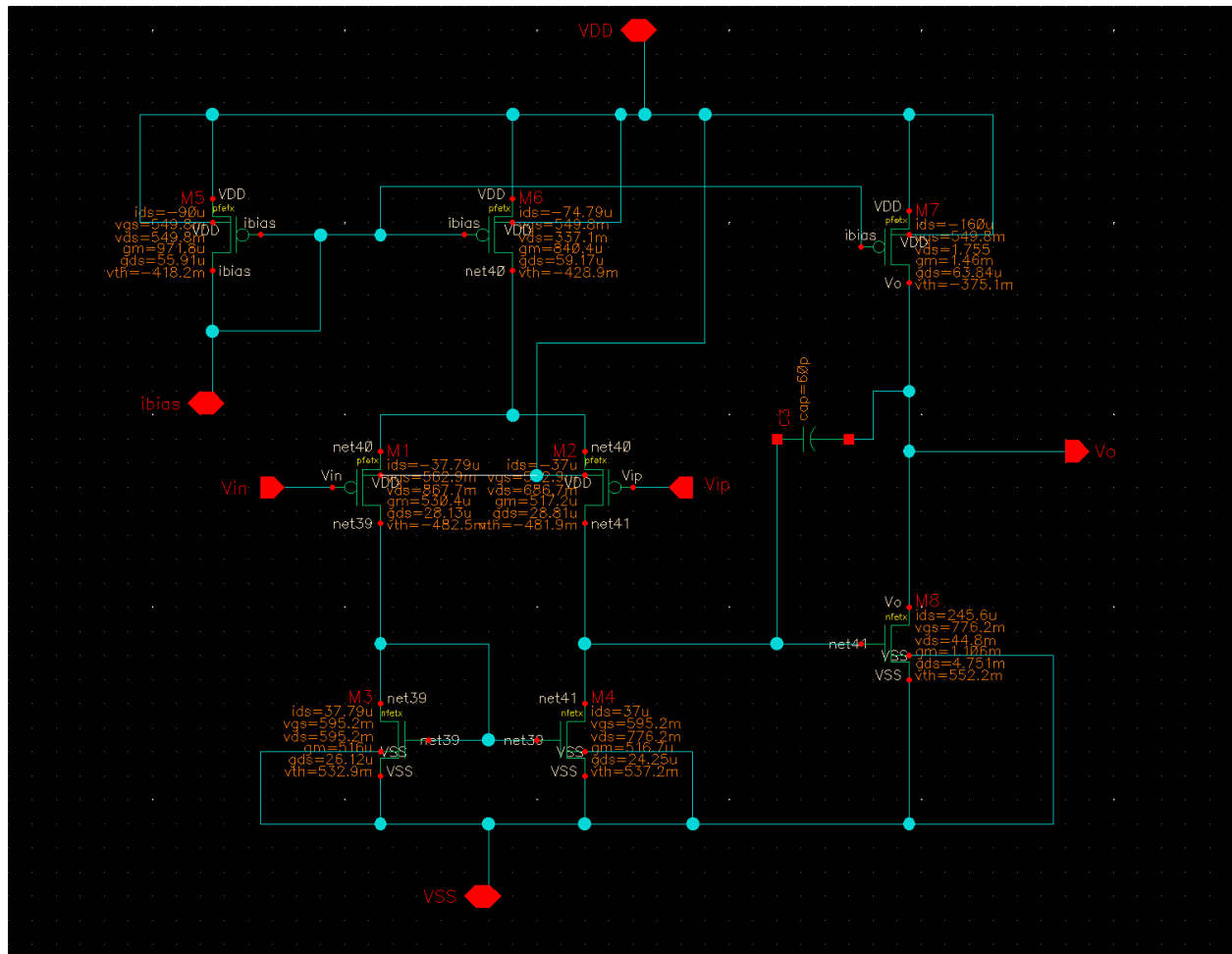
5. Extract the worst & best case 3σ statistical corners for A_{v0} , GBW, PM(Refer [3]) and use the option shown in Fig 3.

Q6. Once you have the corners generated from the Monte Carlo Simulations. Run DC simulation for each corner and note down the DC operating points(I_d , V_{gs} , V_{ds} , V_{th} , g_m , region) for the devices and the DC node voltages

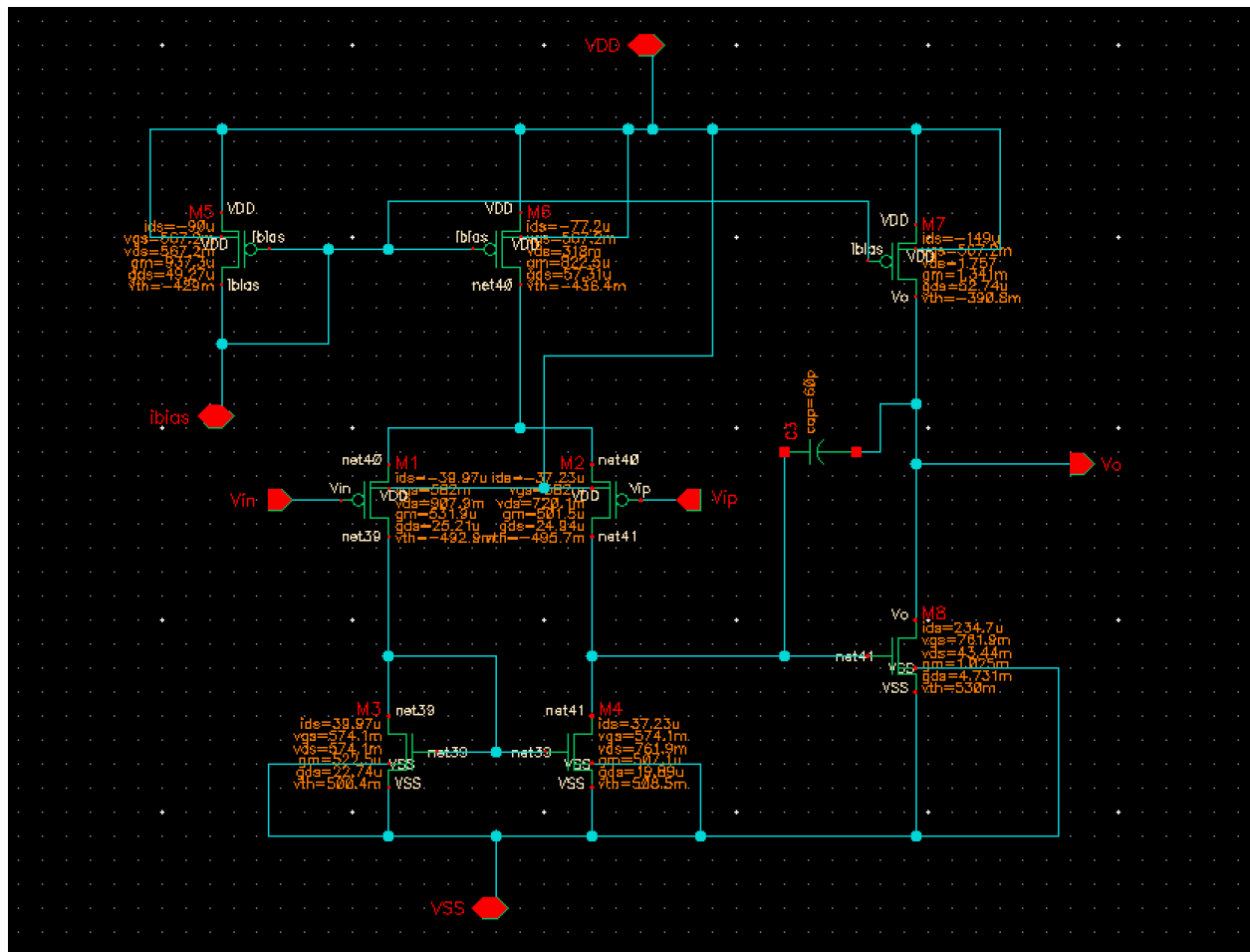
1.Phase-Margin-Low



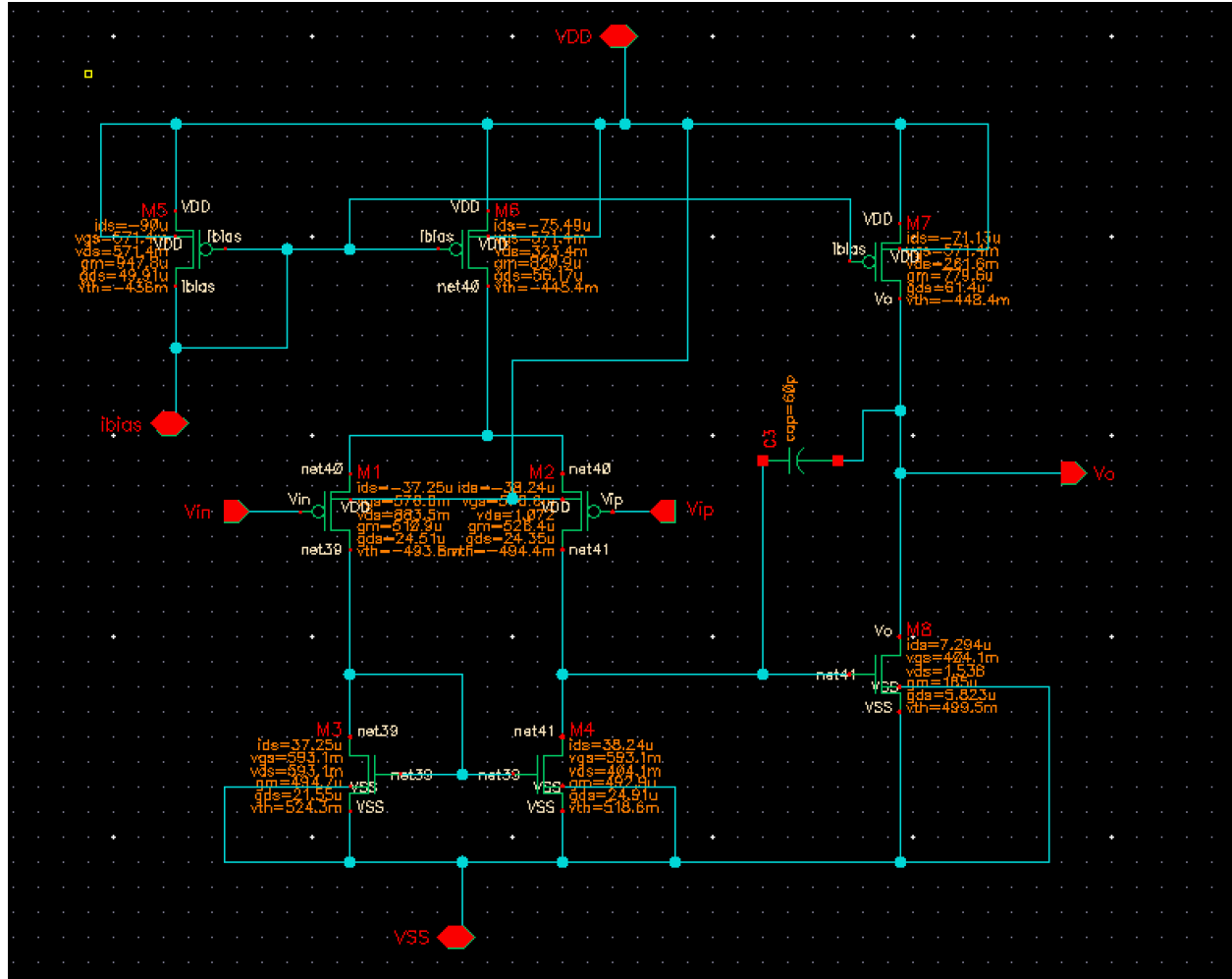
2.Phase_Margin_H



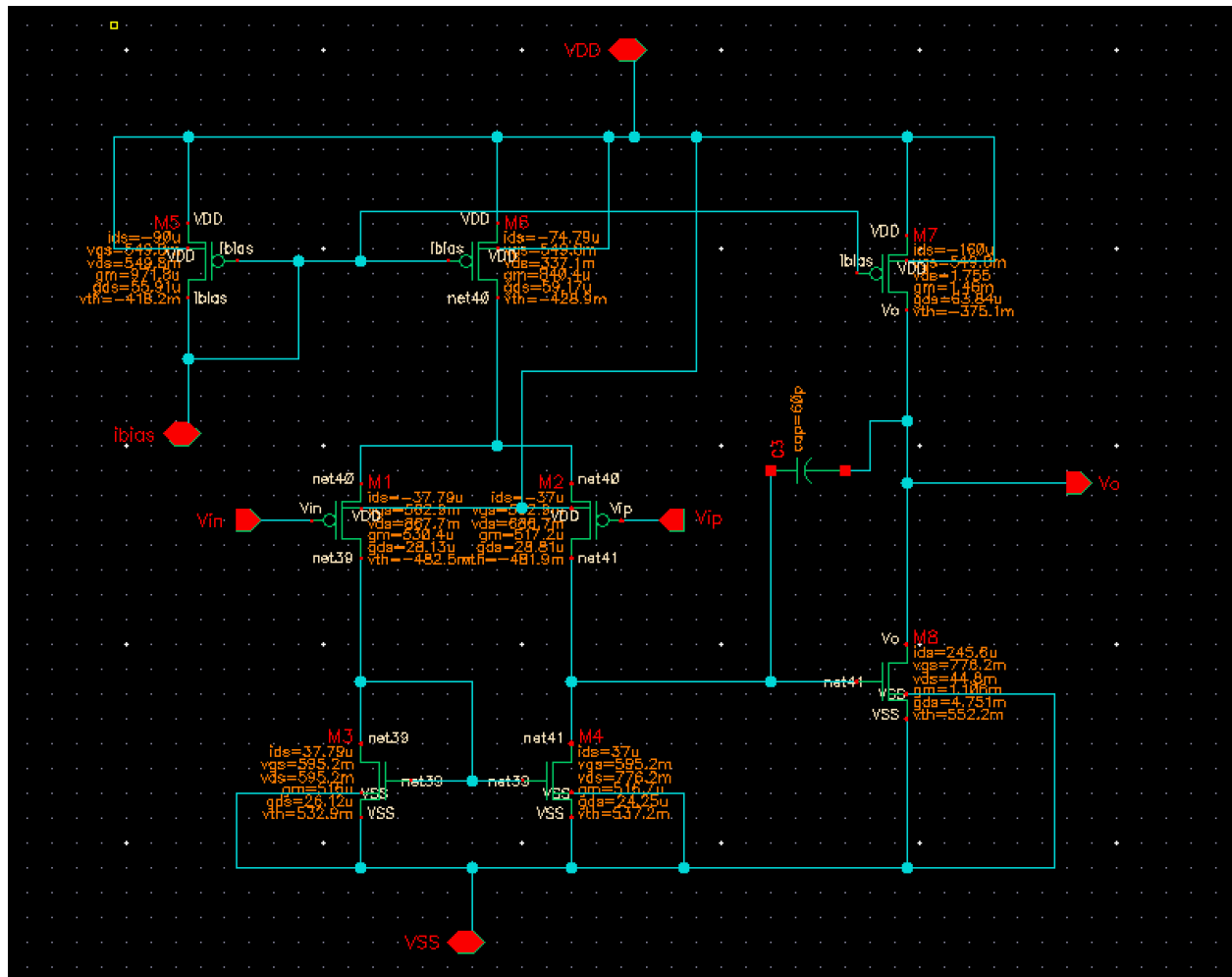
3.GBW_L



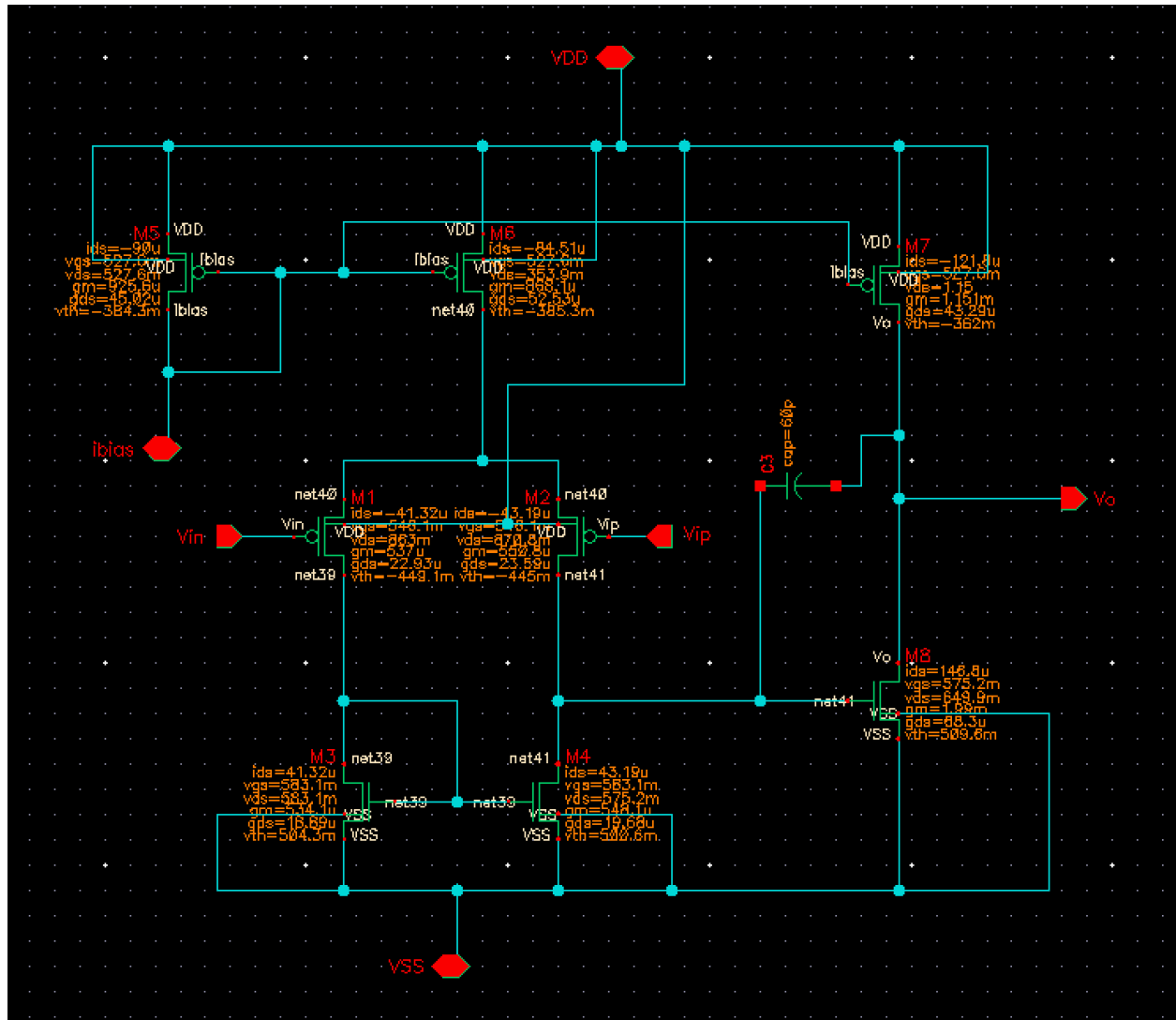
4.GBW_H:



5..Gain_L:



6.Gain_H:



Q7. Based on these observations explain why the specs (A_{v0} , PM, GBW) changes in their respective worst and best corners and ensure that the worst case 3σ is meeting the spec for A_{v0} , GBW, PM. (Show histograms)

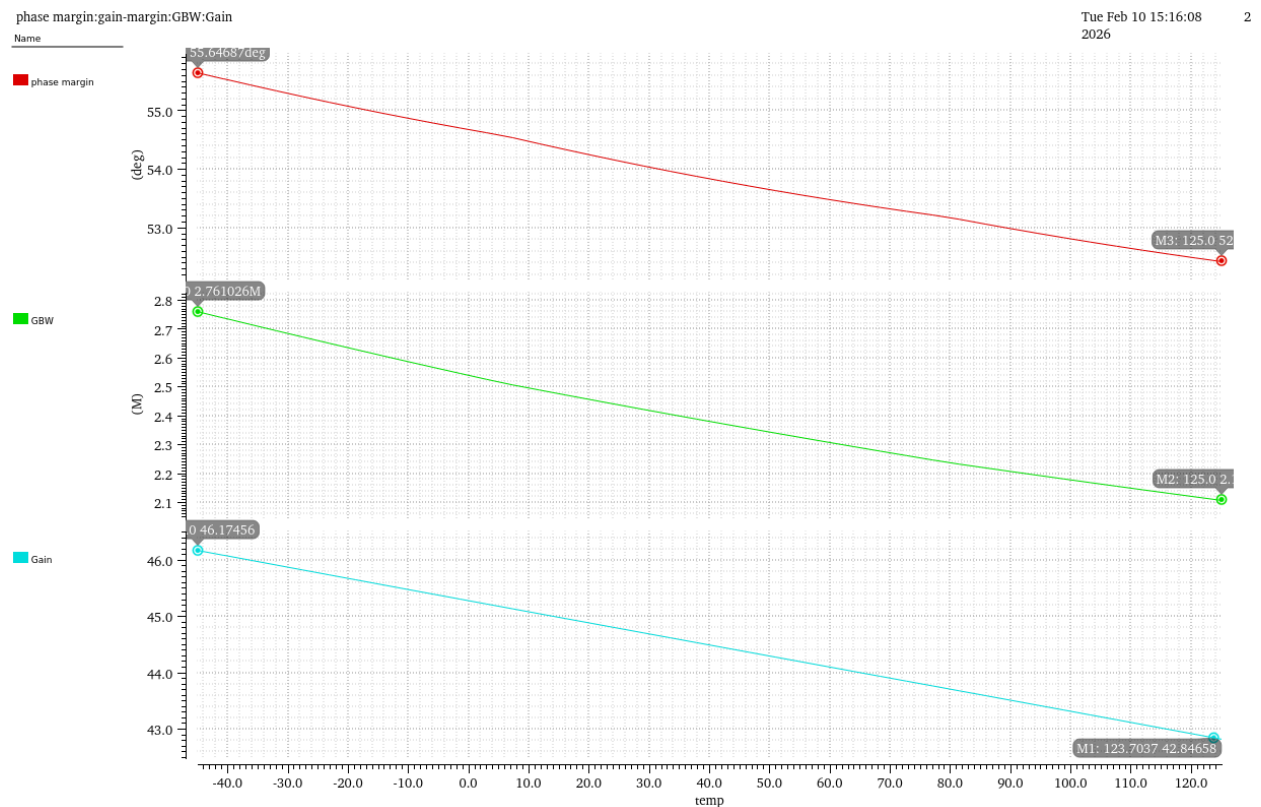
The specs A_{v0} (DC Gain), PM (Phase Margin), and GBW (Gain-Bandwidth Product) change in their respective worst and best corners because of process variations and device mismatch introduced during manufacturing and in MC simulations. These variations affect transistor parameters such as threshold voltage, mobility, and channel length, which in turn influence the operating points (like I_d , V_{gs} , V_{ds} , g_m) of the devices in the op-amp circuit. In the worst corners, unfavorable shifts in these parameters degrade the performance, resulting in reduced gain, bandwidth, or phase margin, while in the best corners, optimal device parameters enhance these specs. This is why

running simulations across corners is necessary—to ensure the design is robust and specification-compliant under all expected variations

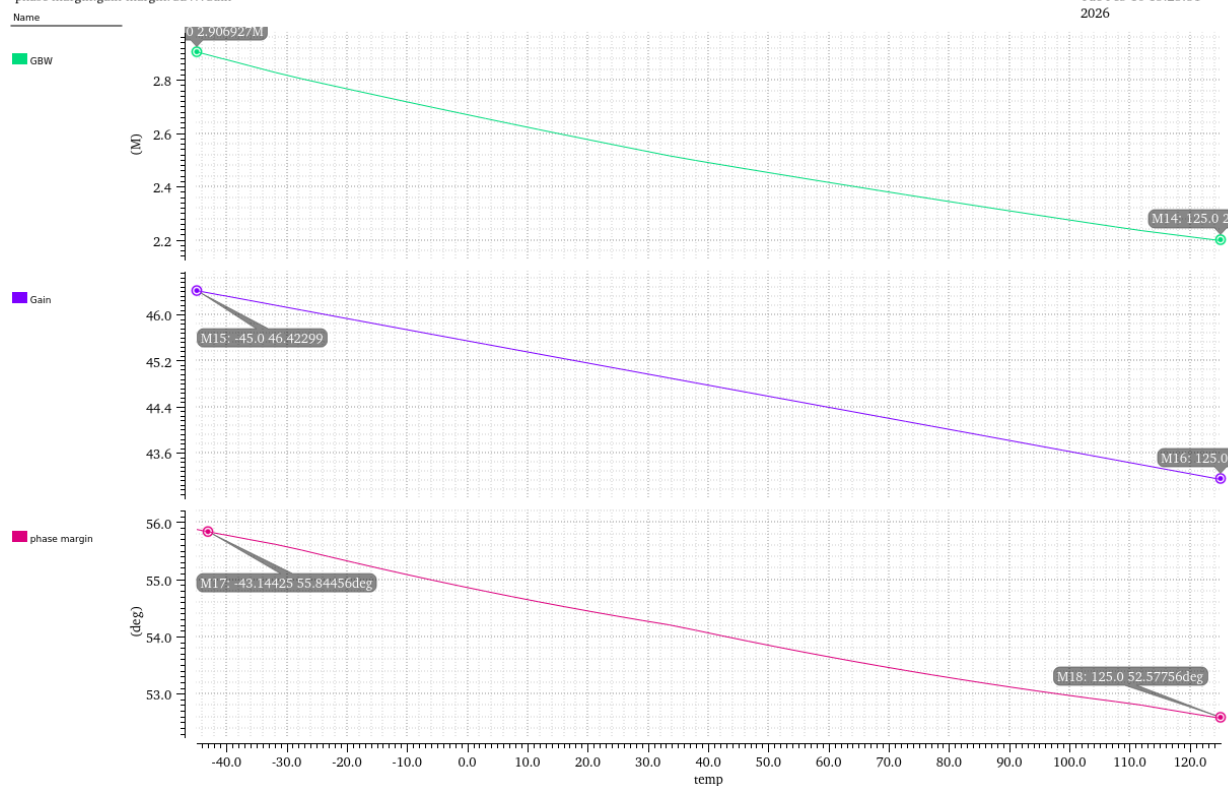
10. Using the corners created, run a cross sweep simulation in ADEXL with $\pm 10\%$ voltage variation,

-40 C < temperature < 125 C and report the worst case DC Gain, GBW and Phase Margin (you don't have to adjust the design if the specs are not met). Provide one plot for each parameter capturing the temp variation, supply variation and the corner and mark the worst case.

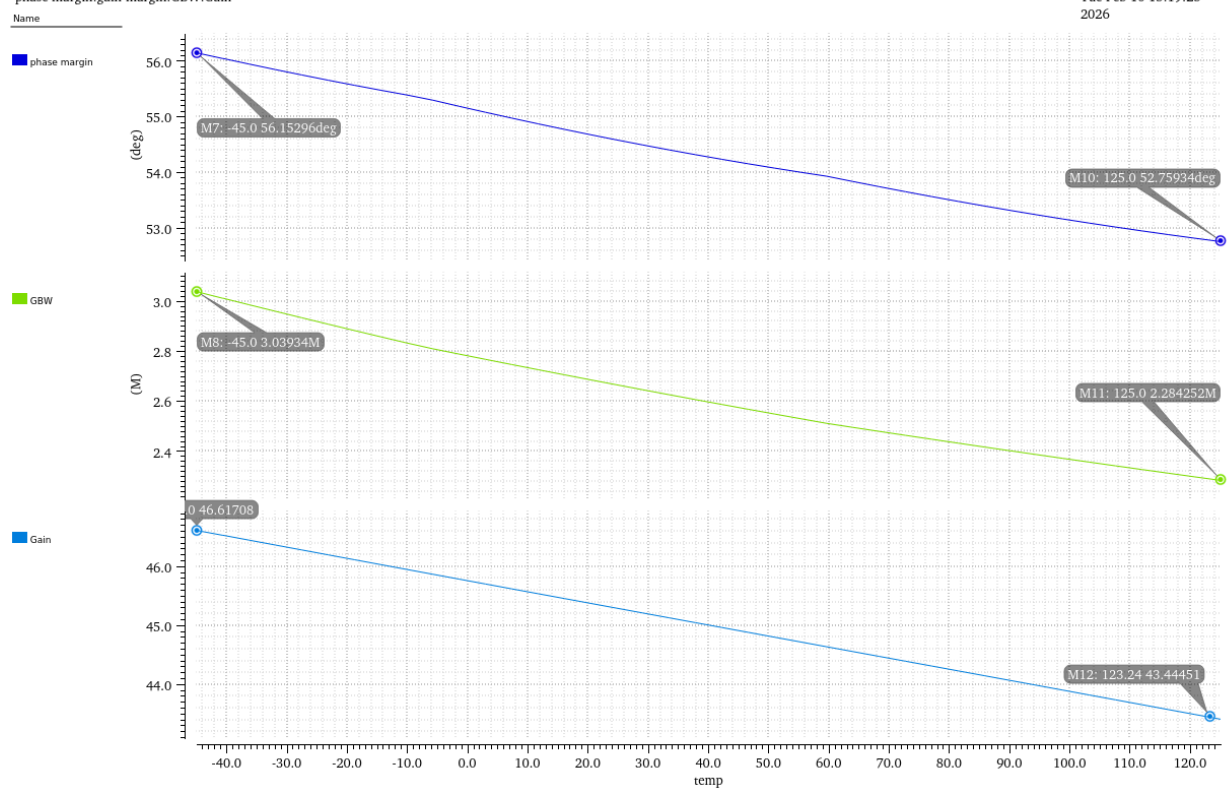
1.FOR-10%:



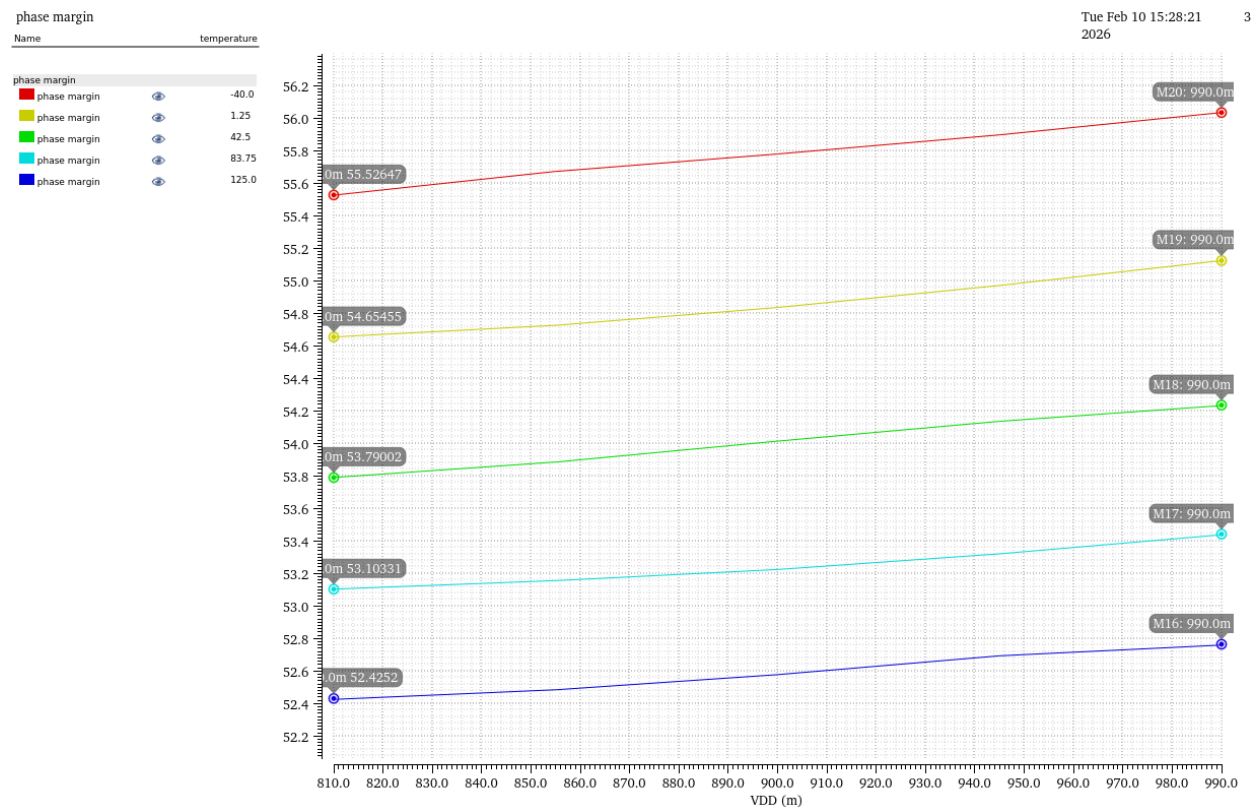
2.FOR NORMAL:



3.FOR +10%

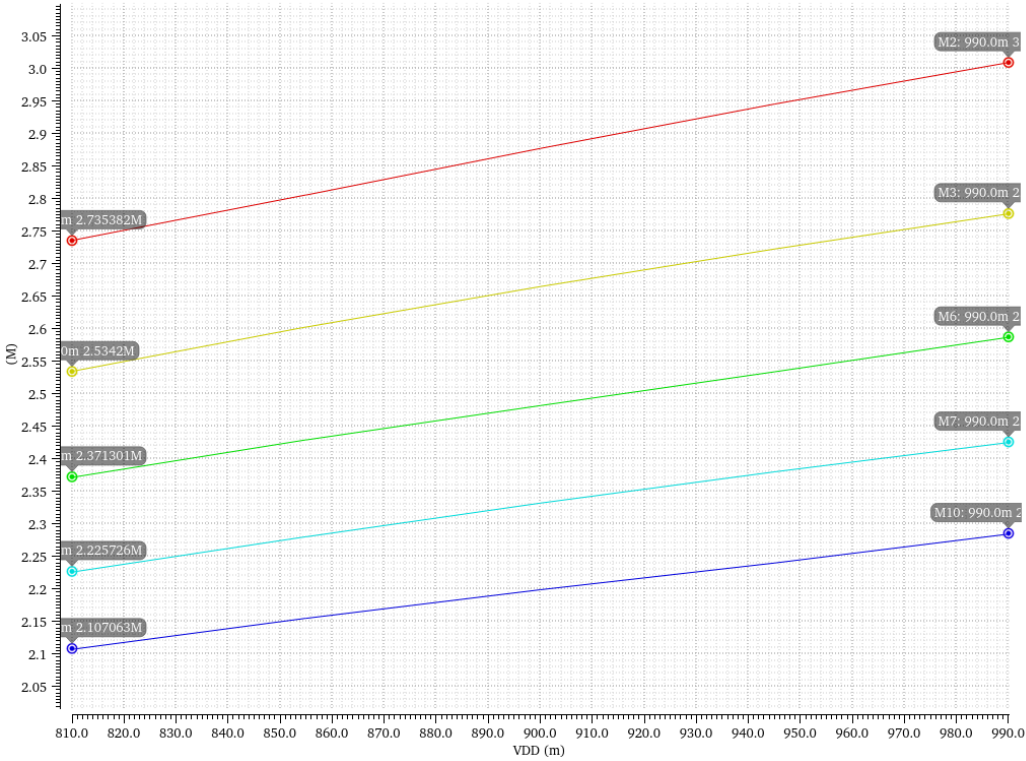


Phase_margin:



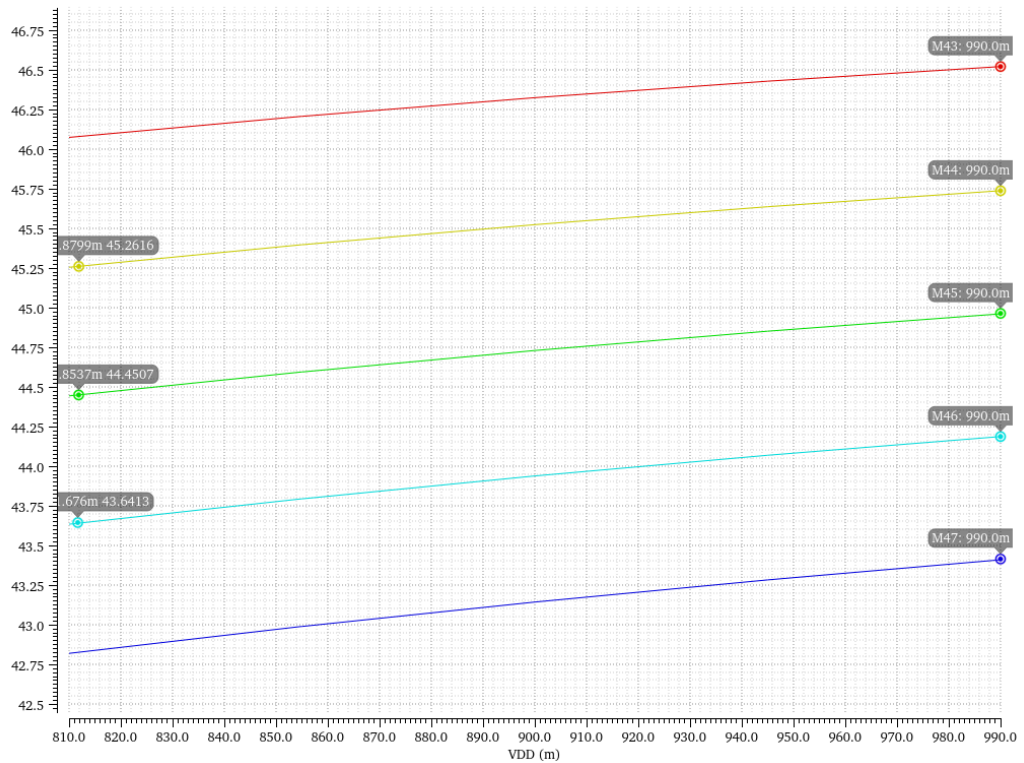
GBW:

GBW		
GBW		-40.0
GBW		1.25
GBW		42.5
GBW		83.75
GBW		125.0



Gain:

Gain	-40.0
Gain	1.25
Gain	42.5
Gain	83.75
Gain	125.0



11. Comment on the impact of process variations and mismatch on each parameter and your overall learning from this lab.

The manual highlights learning outcomes regarding process variation and mismatch. These factors can cause significant spread in op-amp parameters, e.g., reducing DC gain, bandwidth, or phase margin. MC simulations and corner analyses help ensure robust design, confirming specs are met even in worst-case manufacturing and environmental scenarios. Understanding these effects is crucial for designing reliable analog circuits